

Data management project

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Type of project: use of a noSQL tool (i.e. MongoDB)

Datasets:

Spotify Most Streamed Songs

(<https://www.kaggle.com/datasets/abdulszz/spotify-most-streamed-songs>)

This dataset contains comprehensive information on some of the most streamed songs on Spotify, enriched with additional insights from other popular streaming platforms like Apple Music, Deezer, and Shazam. It is ideal for music analysts, data scientists, and machine learning enthusiasts who are interested in exploring trends and characteristics of popular music tracks.

Spotify Tracks Genre

(<https://www.kaggle.com/datasets/thedevastator/spotify-tracks-genre-dataset>)

This dataset provides comprehensive information about Spotify tracks encompassing a diverse collection of 125 genres.

Metacritic Scores for Games, Movies, TV & Music (only music.csv)

(<https://www.kaggle.com/datasets/patk1e/metacritic-scores-for-games-movies-tv-and-music>)

Meta- & User scores for all categories on metacritic.com.

Description of the work: our goal would be to implement a web site. A user will be able to register on the website and subsequently log in. On the homepage, they will find data related to the dataset of the most streamed songs on Spotify and will be able to select artists or tracks to add to their favorites list (like a playlist).

Additionally, they will be able to access the website's archive, which contains all the data from the music.csv and tracks genre datasets. This archive will feature a list of authors along with the albums they have written (including a summary, user score, and metascore) and their songs (with relevant details and parameters). Finally, the user will be able to visit a section dedicated to analyses and trends based on the selected datasets, including trends, parameter-based analyses, and album score evaluations within the dataset.

How our database will be structured:

We have decided to divide our database into three main collections:

1. Users Collection – This will store user data along with their selected favorite artists and tracks.
2. Most Streamed Songs Collection – This will contain data from the "most streamed songs" dataset. Including this dataset allows us to use it for visualization purposes and later for analysis, enabling comparisons with the third dataset.
3. Integrated Music Collection – This will merge the remaining two datasets. Our idea is to integrate them by linking artists, expanding the music.csv dataset with tracks from the Spotify tracks genre dataset, along with their respective parameters. This way, we will have a unified dataset that includes not only an artist's albums and scores but also all the tracks they have released, complete with detailed information about each song.

For data consistency, we have chosen datasets referring to the year 2023.

This approach should represent a real-world scenario, where a website is accessible to users and allows them to perform specific actions: searching, viewing, understanding analyses and trends, and creating a list of favorite artists or tracks.